Exp 16 : SIMULATION AND ANALYSIS OF UNIFORM AND NON-UNIFORM QUANTIZATION USING MATLAB

Objectives :

i) To implement and analyze uniform and non-uniform quantization techniques for an analog signal using MATLAB by converting the continuous-amplitude signal into discrete quantization levels.

ii) To compare the performance of uniform and non-uniform quantization by analyzing their quantized waveforms and quantization errors, and evaluate their effectiveness in representing signals with different amplitude variations.

Code :

```
clc;
clear;
close all;

% Input Signal
t = 0:0.001:1;
x = sin(2*pi*5*t);

% Uniform Quantization (8 Levels)
L = 8;
x_uni = round((x+1)*(L-1)/2);
x_uni = (x_uni*2/(L-1))-1;

% Non-Uniform Quantization (mu-law)
mu = 255;
x_comp = sign(x).*log(1+mu*abs(x))/log(1+mu);
x_non = round((x_comp+1)*(L-1)/2);
x_non = (x_non*2/(L-1))-1;

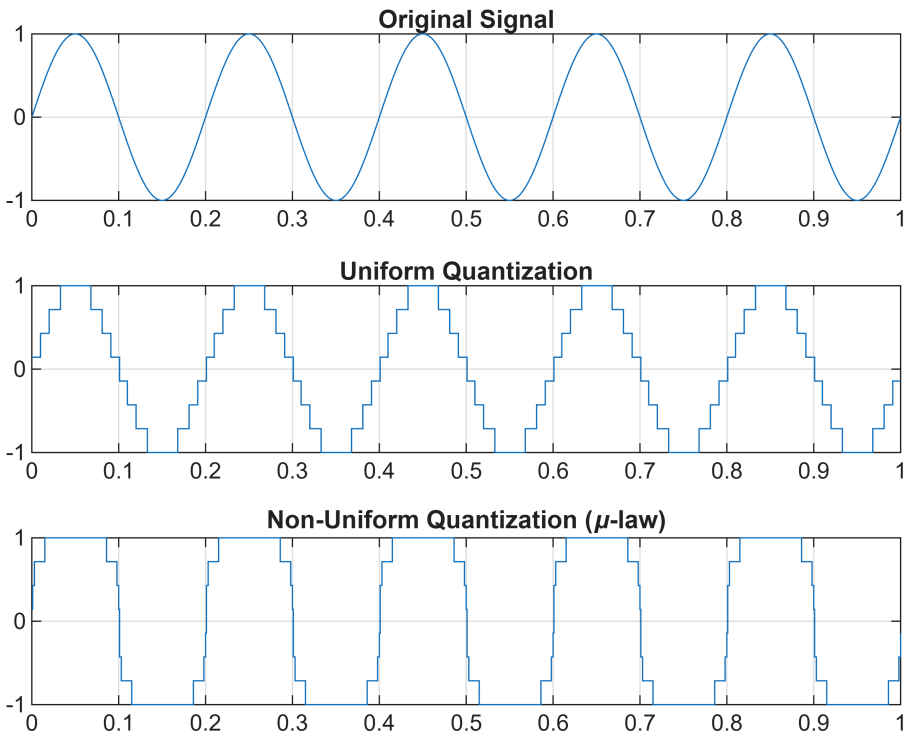
% Plot Results
figure;

subplot(3,1,1)
plot(t,x)
title('Original Signal')
grid on

subplot(3,1,2)
stairs(t,x_uni)
```

```
title('Uniform Quantization')
grid on

subplot(3,1,3)
stairs(t,x_non)
title('Non-Uniform Quantization (\mu-law)')
grid on
```



Result :

i) The uniform and non-uniform quantization techniques were successfully implemented in MATLAB, and the corresponding quantized signals and quantization error waveforms were obtained. ii) The performance comparison showed that non-uniform quantization produced lower quantization error for low-amplitude signals than uniform quantization, resulting in improved signal representation and better overall quantization efficiency.